SDUHSD CAREER TECHNICAL EDUCATION

Human-Computer Interaction (HCI) Design

Campus: Canyon Crest Academy
CTE Industry Sector: Information and Communication Technologies
CTE Pathway: Software and Systems Development
Course Sequence: Capstone
Prerequisites: Computer Science Foundations or Computer Game Design (Completion of CTE Pathway)
OR
AP Computer Science Principles (would not count for CTE Pathway completion)
10 credits, Practical Art or Elective, UC Approved Area G
Recommended Pathway Sequence:

Course 1 (Concentrator)	Course 2 (Capstone)	Course 3	Enrichment
Computer Science Foundations OR Computer Game Design	Human-Computer Interaction (HCI) Design		Machine Learning AP Computer Science Principles AP Computer Science A

Course Description:
Human-Computer Interaction introduces students to the principles of designing effective, accessible, and user-centered digital products. Students explore UI/UX design, software development processes, and collaborative product design using industry-standard tools. Through hands-on projects, students research user needs, create wireframes and prototypes, and build interactive web applications. Topics include design thinking, usability testing, accessibility, version control, APIs, databases, and cloud deployment. The course emphasizes teamwork, iteration, and real-world problem solving.

Course Syllabi:
See teacher websites

Grading:
Grading in this course is standards-based and emphasizes demonstrated understanding, skill development, and thoughtful application of concepts rather than point accumulation. Students are evaluated through a combination of projects, design artifacts, written reflections, and assessments that reflect real-world workflows in computing and human-centered design, with an emphasis on both process and product, including research, iteration, feedback, and revision. Larger projects and assessments carry more weight than daily practice, while formative work is used to support

learning and improvement rather than penalize mistakes. Overall, grades are intended to clearly communicate progress and mastery of course learning objectives and to reflect each student's level of understanding by the end of the course.

Work Load:
(High, Medium, Low)
Class Setting:
This relates directly to how much the student is expected to work independently or in a group requiring self-initiated work, problem solving, etc versus a highly guided and structured class
(Highly Independent, Moderately Independent, Structured)

*Note: Each year all CTE courses are reviewed by industry professionals within the pathway to ensure the most current, relevant experience for our students.

Textbooks or Supplemental Materials for this course:
This course does not rely on a traditional textbook. Instead, students use a curated set of digital readings, videos, design case studies, and instructor-created materials. Supplemental resources include usability and accessibility guidelines, short articles on human-centered design, and example products for critique and analysis. Materials are updated regularly to reflect current tools, design practices, and real-world applications of HCI and UX.

Software, program knowledge specific to this course:
Students work with industry-relevant tools used in design and front-end development, including Figma for wireframing and prototyping, GitHub for version control and collaboration, and modern web technologies such as HTML, CSS, JavaScript, and React. The course also introduces cloud-based tools and APIs for simple backend functionality, along with usability testing and accessibility evaluation tools. Emphasis is placed on understanding design workflows, collaboration practices, and iterative development rather than mastery of a single platform.

Units of Study, Large Projects, Work-based Learning Connections.
Units progress from HCI foundations and UX principles to design thinking, needfinding, prototyping, and front-end implementation. Students complete multiple applied projects, including user research studies, low- and high-fidelity prototypes, and a culminating team-based capstone project in which they design, build, and present a functional web application. Work-based learning connections include industry-style critiques, peer and stakeholder feedback, collaborative team roles, and exposure to real-world design constraints such as accessibility, usability, and ethical considerations in technology design.

Industry Connections:
University of California, San Diego - Professor - Philip Guo
University of California, San Diego - PhD Graduate Student - Irene Hou

The following Model Curriculum Standards connected to this course:

Standards for Career Ready Practice
1. Apply appropriate skills and technical knowledge. 2. Communicate clearly, effectively, and with reason. 4. Apply technology to enhance productivity. 5. Utilize critical thinking to make sense of problems and persevere in solving them. 8. Model integrity, ethical leadership, and effective management. 9. Work productively in teams while integrating cultural and global competence. 10. Demonstrate creativity and innovation.

Framework Concept	Software and Systems Development Standards
C1.0 Identify and apply the systems development process.	• C1.1 Identify the phases of the systems development life cycle, including analysis, design, programming, testing, implementation, maintenance, and improvement. • C1.2 Identify and describe models of systems development, systems development life cycle (SDLC), and agile computing. • C1.4 Work as a member of, and within the scope and boundaries of, a development project team. • C1.5 Track development project milestones using the concept of versions.
C2.0 Define and analyze systems and software requirements.	• C2.3 Develop strategies that target the specific needs and desires of the customer. • C2.4 Analyze customers' needs for development.
C3.0 Create effective interfaces between humans and technology.	• C3.1 Describe and apply the basic process of input, processing, and output. • C3.2 Design effective and intuitive interfaces using knowledge of cognitive, physical, and social interactions. • C3.3 Support methods of accessibility for all potential users, including users with disabilities and non-English-speaking users.
C4.0 Develop software using programming languages.	• C4.6 Use proper programming language syntax. • C4.7 Use various data structures, arrays, objects, files, and databases. • C4.9 Create programs using control structures, procedures, functions, parameters, variables, error recovery, and recursion. • C4.11 Document development work for various audiences, such as comments for other programmers, and manuals for users.
C5.0 Test, debug, and improve software development work.	• C5.3 Use strategies to optimize code for improved performance. • C5.4 Test software and projects. • C5.5 Evaluate results against initial requirements.
C7.0 Develop Web and online projects.	• C7.5 Create an online project, Web-based business, and e-portfolio.
C8.0 Develop databases.	• C8.1 Describe the critical function of databases in modern organizations. • C8.2 Identify and use the basic structures of databases, fields, records, tables, and views. • C8.5 Use queries to extract and manipulate data (select queries, action queries). • C8.6 Develop databases that are properly normalized using appropriate schemas.

	• C8.7 Export and import data to and from other applications and a database recognizing the limitations and challenges inherent in the process.